

Project Safe Haven CIS-212 Gramel

1. User interface (UI) for Suicide Prevention

The app should have a user-friendly interface where users can:

- * Inputs Student Mental Health Issues
- * Track Counseling
- * View their Progress
- * Set Personal Goals
- A. Mental health Counseling
- B. Group Counseling
- C. Therapy
- D. 24 Hour Hotline

E. Progress and Analytics

- * Progress Charts and Graphs: Visualize your progress.
- * Insight: Provide Analytical (Improvements).

2. Goal Setting

- *Set Strength Goals: users can set future goals and accomplishments
- *Progress Notifications: notify users when goals are done.

3. List Essential Features

- *User Registration and Authentication
- *Mood Tracking
- *Daily Journaling
- *Accessibility and Integration with Resources
- *Privacy and Security
- *User Feedback and Contact Options
- *Push Notifications

4. Steps to Build And Validate Your MVP

1. **Prototype & design**

Technology Stack

A. Frontend (Mobile Développement):

- *React Native for building cross-platform mobile apps (IOS and Android).

Use libraries like react-native-chart-kit to visualize strength progress and workout history in chart.

B. Backend(Server-Side Development):

- *Firebase for updates and user authentication.
- *Node.js or Python Flask/Django for server-side logic to handle all data.

5. Testing and Development

- *Test app to ensure data is correctly logged and displayed (GUI) and sustainability.
- *Use Firebase Emulator Suite or other tools for backend testing.
- *Deploy app to Google Store or Apple App Developers after extensive testing.

6. Scale Up:

Once App is launched our team will be adding more features advanced mood analysis, community features, integration of wearable devices, gather insights from users and create IDE.

CONCLUSION

Our Team proposes using Kanban to manage the development of Safe Haven, mental health app tailored to meet the needs of Belhaven students. This proposal outlines the need for this change, why Kanban was selected as the project management solution, the risks involved and their mitigations, and a plan for rolling out the app and the Kanban framework. We look forward to your feedback on this proposal and to enacting the plan once we've addressed any concerns you may have. At Safe Haven all Staff members are our first responders. Counseling Staff 24 hrs.

1. **Comprehensive Resource Access:** The app will include a clear, organized list of mental health services and tools available to students.
2. **Anonymous Support Options:** Features such as anonymous chat and peer support will reduce stigma and encourage engagement.
3. **Self-Help Tools:** Guided meditation, journaling, and other self-help features will provide consistent, on-demand support for students' mental health journeys.
4. **Centralized Platform:** Safe Haven will integrate existing campus resources and external mental health tools into a single, user-friendly app.

By using Kanban as our project management framework, we can develop the app efficiently while maintaining high quality.

BEGIN MentalHealthApp

// Function: RegisterUser

// Purpose: Register a new user in the app

Input: username, password, email

Output: userID

```
Procedure RegisterUser(username, password, email):  
    IF username, password, email ARE valid THEN  
        userID = CreateNewUser(username, password, email)  
        RETURN userID  
    ELSE  
        RETURN "Invalid input"  
    END IF  
END Procedure
```

// Function: Login

// Purpose: Authenticate user login

Input: username, password

Output: Boolean

```
Procedure Login(username, password):  
    IF ValidateUserCredentials(username, password) THEN  
        RETURN True  
    ELSE  
        RETURN False  
    END IF  
END Procedure
```

// Function: TrackMood

// Purpose: Save user mood log

Input: userID, mood, date

Output: status

Procedure TrackMood(userID, mood, date):

 IF userID is valid THEN

 SaveMoodEntry(userID, mood, date)

 RETURN "Mood entry saved"

 ELSE

 RETURN "Invalid user ID"

 END IF

END Procedure

// Function: JournalEntry

// Purpose: Save user journal entry

Input: userID, entry text, date

Output: status

Procedure JournalEntry(userID, entry_text, date):

 IF userID is valid THEN

 SaveJournalEntry(userID, entry_text, date)

 RETURN "Journal entry saved"

 ELSE

 RETURN "Invalid user ID"

```
END IF  
END Procedure
```

```
// Function: SetReminder
```

```
// Purpose: Set a reminder for the user
```

```
Input: userID, reminder text, time
```

```
Output: status
```

```
Procedure SetReminder(userID, reminder_text, time):
```

```
    IF userID is valid THEN
```

```
        ScheduleReminder(userID, reminder_text, time)
```

```
        RETURN "Reminder set"
```

```
    ELSE
```

```
        RETURN "Invalid user ID"
```

```
    END IF
```

```
END Procedure
```

```
// Function: GetResources
```

```
// Purpose: Provide mental health resources
```

```
Output: list of resources
```

```
Procedure GetResources():
```

```
    resources = FetchMentalHealthResources()
```

```
    RETURN resources
```

END Procedure

// Main Controller Logic

Procedure Main():

 WHILE App IS active

 DISPLAY "Welcome to the Mental Health App"

 DISPLAY Options: "1. Register, 2. Login, 3. Exit"

 user_choice = GetUserInput()

 IF user_choice == 1

 DISPLAY "Enter Username, Password, Email"

 username, password, email = GetUserInput()

 status = RegisterUser(username, password, email)

 DISPLAY status

 ELSE IF user_choice == 2

 DISPLAY "Enter Username, Password"

 username, password = GetUserInput()

 IF Login(username, password) == True

 userID = GetUserID(username)

 DISPLAY "Login Successful"

DISPLAY Options: "1. Track Mood, 2. Add Journal, 3. Set Reminder, 4. Get Resources, 5. Logout"

user_action = GetUserInput()

IF user_action == 1

DISPLAY "Enter Mood and Date"

mood, date = GetUserInput()

status = TrackMood(userID, mood, date)

DISPLAY status

ELSE IF user_action == 2

DISPLAY "Enter Journal Entry and Date"

entry_text, date = GetUserInput()

status = JournalEntry(userID, entry_text, date)

DISPLAY status

ELSE IF user_action == 3

DISPLAY "Enter Reminder Text and Time"

reminder_text, time = GetUserInput()

status = SetReminder(userID, reminder_text,
time)

DISPLAY status

ELSE IF user_action == 4

resources = GetResources()

```
        DISPLAY resources
    END IF
ELSE
    DISPLAY "Login Failed"
END IF

ELSE IF user_choice == 3
    DISPLAY "Goodbye!"
    BREAK
END IF

END WHILE

END Procedure
```

END MentalHealthApp

...

Pseudocode provided a simple code with the essential components to store data, user input, error handling, validation, security and more features to come after beta testing complete.

Start date: 03-22-2025

End Date: 03-23-2025

